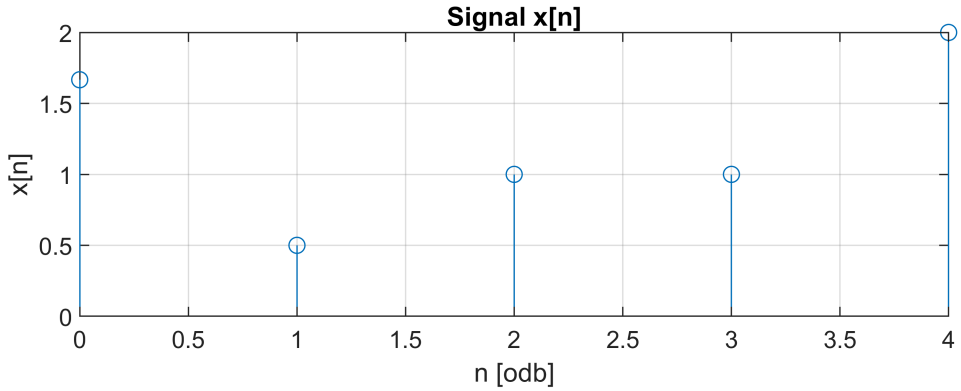
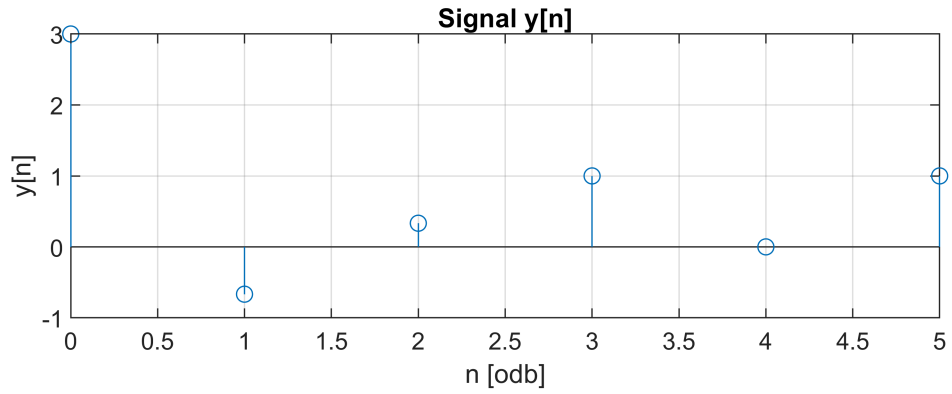
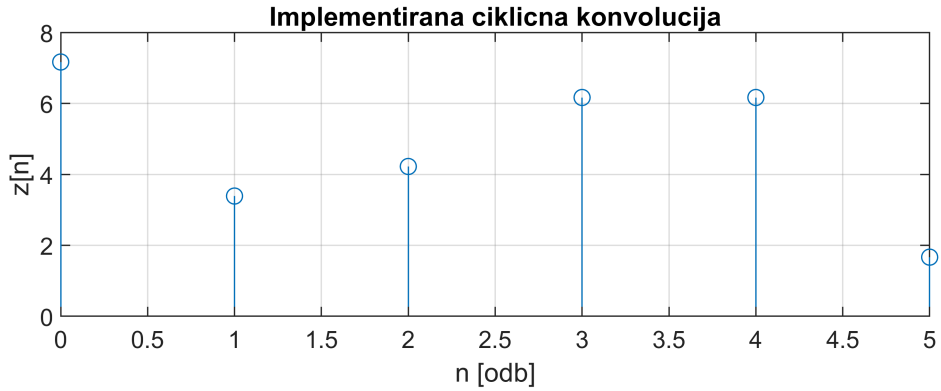
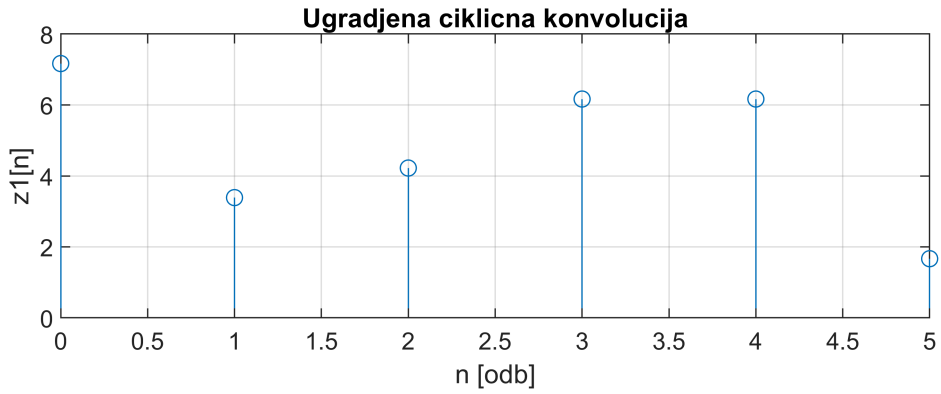
```
clear all; close all; clc;

P = mod(537, 4) + 4;
%---------------------------------------------
n1 = 0 : floor(P/3) - 1;
n2 = floor(P/3) : floor(2*P/3) - 1;
n3 = floor(2*P/3) : P-1;

x1 = 2*sin(n1) + P/3;
x2 = 2.^(n2-2);
x3 = n3-2;

nx = [n1, n2, n3];
x = [x1, x2, x3];
%----------------------------------------------
n4 = 0 : floor(P/4) - 1;
n5 = floor(P/4) : floor(3*P/4) - 1;
n6 = floor(3*P/4) : P;

y1 = (-1).^n4 + 2;
y2 = n5 - P/3;
y3 = mod(n6, 2);

ny = [n4, n5, n6];
y = [y1, y2, y3];
%---------------------------------------------
figure(1); %crtanje grafika za signale x i y

subplot(2, 1, 1);
stem(nx, x);
xlabel('n [odb]'); ylabel('x[n]'); title('Signal x[n]'); grid on;

subplot(2, 1, 2);
stem(ny, y);
xlabel('n [odb]'); ylabel('y[n]'); title('Signal y[n]'); grid on;
```





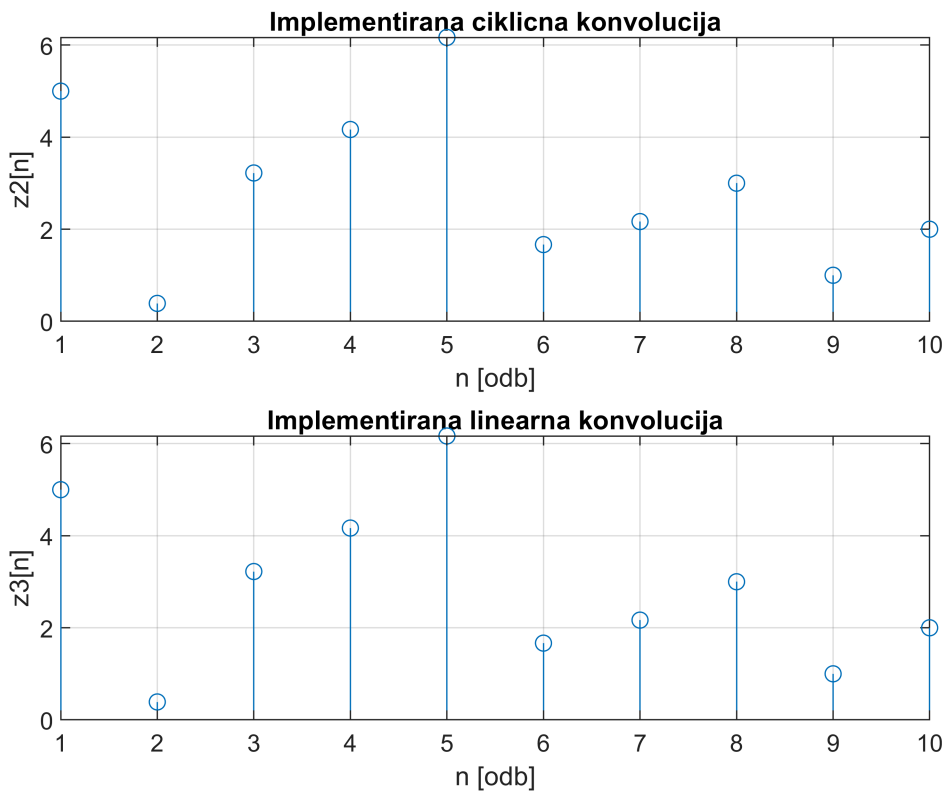

```
%-----------------------------------------------
N = P+1; %broj tacaka u kojima racunamo po zadatku

xz = zeros(1, N - length(x)); %dopunjavanje nulama do potrebne duzine
yz = zeros(1, N - length(y));
x1 = [x, xz];
y1 = [y, yz];
%-----------------------------------------------
z = zeros(1, N); %implementirana ciklicna konvolucija
for k = 0 : N-1
    z(k+1) = sum(x1.*y1(mod(k : -1 : k-(N-1), N)+ 1));
end

z1 = cconv(x, y, N); %ugradjena ciklicna konvolucija
%-----------------------------------------------
n = 0 : N-1;
figure(2); %crtanje grafika dobijenih konvolucija

subplot(2, 1, 1);
stem(n, z);
xlabel('n [odb]'); ylabel('z[n]'); title('Implementirana ciklicna konvolucija'); grid on;

subplot(2, 1, 2);
stem(n, z1);
xlabel('n [odb]'); ylabel('z1[n]'); title('Ugradjena ciklicna konvolucija'); grid on;
```



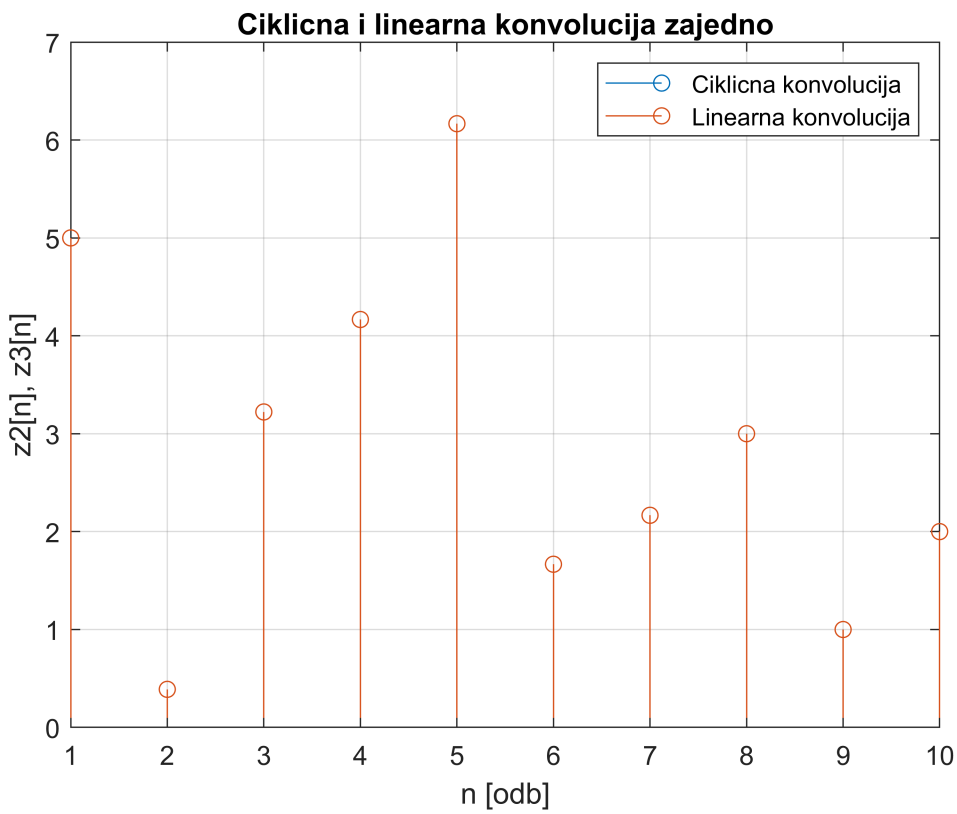



```
%-----------------------------------------------------
% N = length(x)+length(y)-1 => cirkularna je ekvivalentna linearnoj konvoluciji
N1 = length(x) + length(y) - 1;

xz2 = zeros(1, N1 - length(x)); %dopunjavanje nulama do potrebne duzine
yz2 = zeros(1, N1 - length(y));
x2 = [x, xz2];
y2 = [y, yz2];

z2 = zeros(1, N1); %implementirana ciklicna konvolucija
for k = 0 : N1-1
    z2(k+1) = sum(x2.*y2(mod(k : -1 : k-(N1-1), N1)+ 1));
end

z3 = zeros(1, N1); %implementirana linearna konvolucija
for i = 0 : N1 - 1
    sum = 0;
    for j = 0 : length(x) - 1
        if(j <= length(x)-1 && i-j >= 0 && i-j <= length(y)-1)
            sum = sum + x(j+1)*y(i-j+1);
        end
    end
    z3(i+1) = sum;
end

z4 = conv(x, y); %ugradjena linearna konvolucija
%-----------------------------------------------------
```

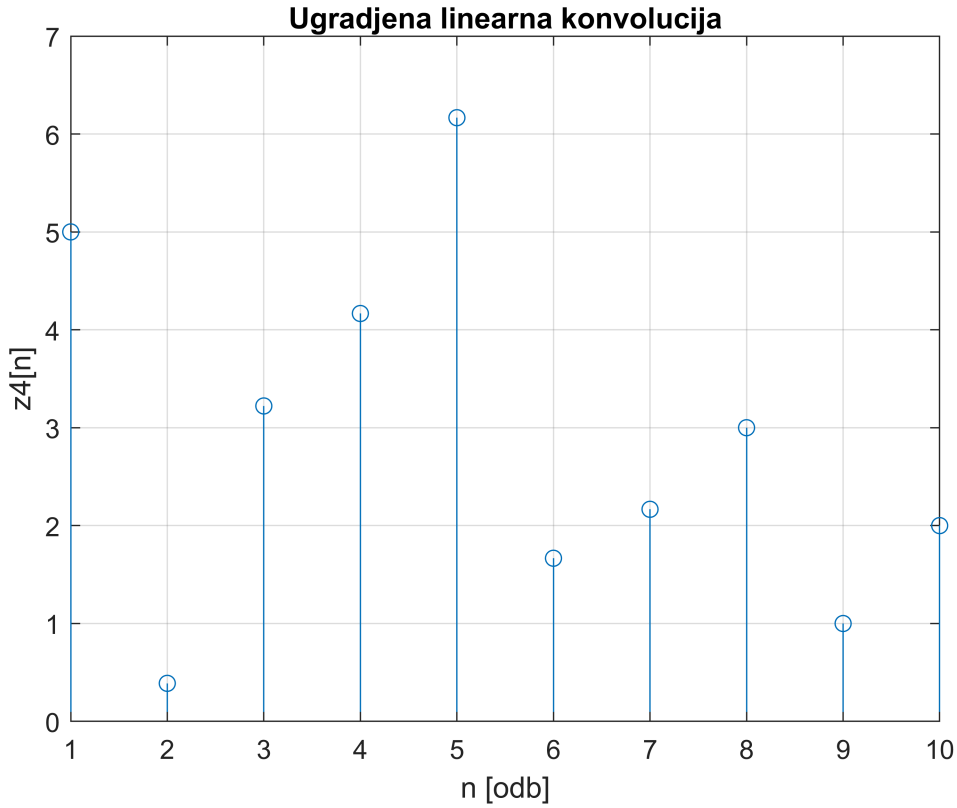
```
figure(3); %crtanje grafika dobijenih konvolucija

subplot(2, 1, 1);
stem(z2);
xlabel('n [odb]'); ylabel('z2[n]'); title('Implementirana ciklicna konvolucija'); grid on;

subplot(2, 1, 2);
stem(z3);
xlabel('n [odb]'); ylabel('z3[n]'); title('Implementirana linearna konvolucija'); grid on;
```




```
figure(4);
stem(z2);
hold on;
stem(z3);
xlabel('n [odb]'); ylabel('z2[n], z3[n]');
title('Ciklicna i linearna konvolucija zajedno'); grid on;
legend('Ciklicna konvolucija', 'Linearna konvolucija');
```




```
figure(5);
stem(z4);
xlabel('n [odb]'); ylabel('z4[n]'); title('Ugradjena linearna konvolucija'); grid on;
```


Ugradjena linearna konvolucija
z4[n]
n [odb]
0
1
2
3
4
5
6
7
1
2
3
4
5
6
7
8
9
10